

Case study

Research question: "Does Annual Household Income Play a Role in Minority's Political Party Identification?"

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Questions to answer:

- Try to explain why this research question might be of interest and to whom.

Usually, individuals identify to political parties regarding multiple parameters, such as economical positions (liberal vs conservative), work-related or ecological propositions, positions regarding culture (open, nationalist, etc.), or positions regarding main religions (catholic, protestant, muslim, jewish). But they can also feel represented by the individuals that are members of the party, if they have similar backgrounds for example.

Minorities are usually less represented in party members, and their religions or culture might not even be a topic of some parties, therefore some parameters like income might play a bigger role in party identification.

This is useful to know for parties who want to better represent a larger part of the country's population, or understand better the profile of their potential voters. It can also be useful for the government in place to know if they can improve their policies for minorities to feel better represented.

- Locate relevant data file and accompanying documentation in the "stata" folder.

The relevant file is located in "UKDA-6970-stata\BSE_2010.dta", along with the documentation in the "mrdoc" folder, especially the "allissue" and "pdf" folders.

- Perform any necessary transformations of data format type.

The stata data file can be open in several ways. We'll work in Python so it can easily be opened and transformed in a dataframe with the pandas library (see attached python code).

- Identify variables of interest and specify their type.
 - Variable = **zq91_1 = N People Household** (Numeric, -2 and -1 are nominal, representing respectively "Refused" and "Don't Know")
 - Variable = **bq9_1 = Party Identification** (Numeric, SPSS measurement level is nominal (each positive number is a different party)).
 - Variable = **zqinc = Annual Household Income**

This variable is nominal.

- Variable = **Weight_trimmedF2FALL5 = Weight for equal comparison** of each ethnicity for analysis of all 5 groups as one population

This variable is numeric, and can be used with the following variable for equal comparison of ethnicities:

- Variable = **ScreenEthQairre = 5 ethnicities surveyed**

This variable is numeric, the SPSS measurement level is scale (each different number is an ethnicity).

- In case a variable is ordinal or nominal, list the possible categories and their meaning.

Value label information for bq9_1 (Party Identification)		Value label information for zqinc (Household Income)	
Value	Label	Value	Label
-2	Refused	-2	Refused
-1	Don't Know	-1	Don't Know
1	None/No	1	0 - 5000
2	Labour	2	5001 - 10000
3	Conservatives	3	10001- 15000
4	Liberal Democrats	4	15001 - 20000
5	Scottish National Party (SNP)	5	20001 - 25000
6	Plaid Cymru	6	25001 - 30000
7	Green Party	7	30001 - 35000
8	United Kingdom Independence Party (UKIP)	8	35001 - 40000
9	British National Party (BNP)	9	41000 - 45000
10	Coalition party/Conservative-Lib dem party	10	45001 - 50000
11	Other	11	50001 - 60000
12	Respect	12	60001 - 70000
15	Not Stated	13	70001 - 80000
		14	80001 or more
		17	Not Stated

Value label information for ScreenEthQairre (5 ethnicities)	
Value	Label
1	Black Caribbean
2	Black African
3	Indian
4	Pakistani
5	Bangladeshi

- Define the analysis required to answer the RQ and explain why you have chosen this particular analysis.

Chi² - categorical (political identification) vs categorical (income range)

Modifying income range in numbers, and grouping by political identification to have a mean income vs political identification (t stat), or linear regression analyses.

If Simple Chi² doesn't yield results, it could be interesting to apply transformations into less categories, for example low, medium and high income, or major/big parties vs small parties.

- List all assumptions and how you addressed them.

Assumption 1: For Party Identification, -2 is considered NaN for our study (refused to answer). We keep "Don't know" as we consider it a valid answer for our research question.

Assumption 2: Similarly, -2 and -1 are considered NaN for Household Income.

Assumption 3: The data can be incomplete in a lot of different ways, so we need to clear out all NaN values from the dataframe before performing analysis.

- List the null and alternative hypothesis.

H0 (Null): Household Income doesn't affect political identification for minorities.

H1: Household income affects political identification.

H1a: Low income is related to identification with populist parties or leftist parties.

- Provide the results of your analysis and answer the research question.

1. Cleaning of the data, analysis of missing data and exploratory sensitivity analysis

Missing value analysis showed a lot of missing values in one of the variables of interest (or values that are not useful for the analysis, such as Refused to answer).

Additional sensitivity analysis showed that even with the provided weight for population representation, there might be an imbalance in representation due to the amount of missing values being uneven between ethnicities.

Our exploratory chi-square test gave **$\chi^2 = 26.43$, with 12 degrees of freedom and $p = 0.0093$** . Under the ordinary test's assumptions, this provides evidence against income and party identification being independent.

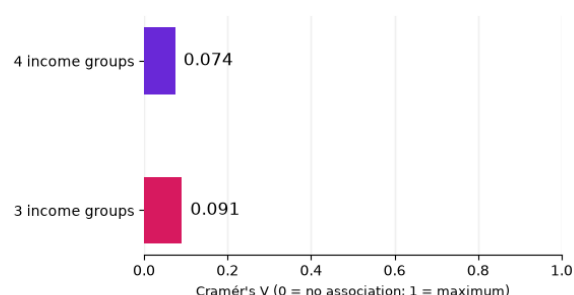
However, **Cramér's V was 0.074**, indicating a small association. Statistical significance does not mean that income strongly explains party identification.

Sensitivity check: two income groupings, the same respondents

Ordinary chi-square results (unweighted)

Grouping	N	Chi-square	df	p-value	Min. expected count
4 income groups	1,597	26.43	12	0.0093	5.21
3 income groups	1,597	26.19	8	0.0010	5.21

Measured association is small

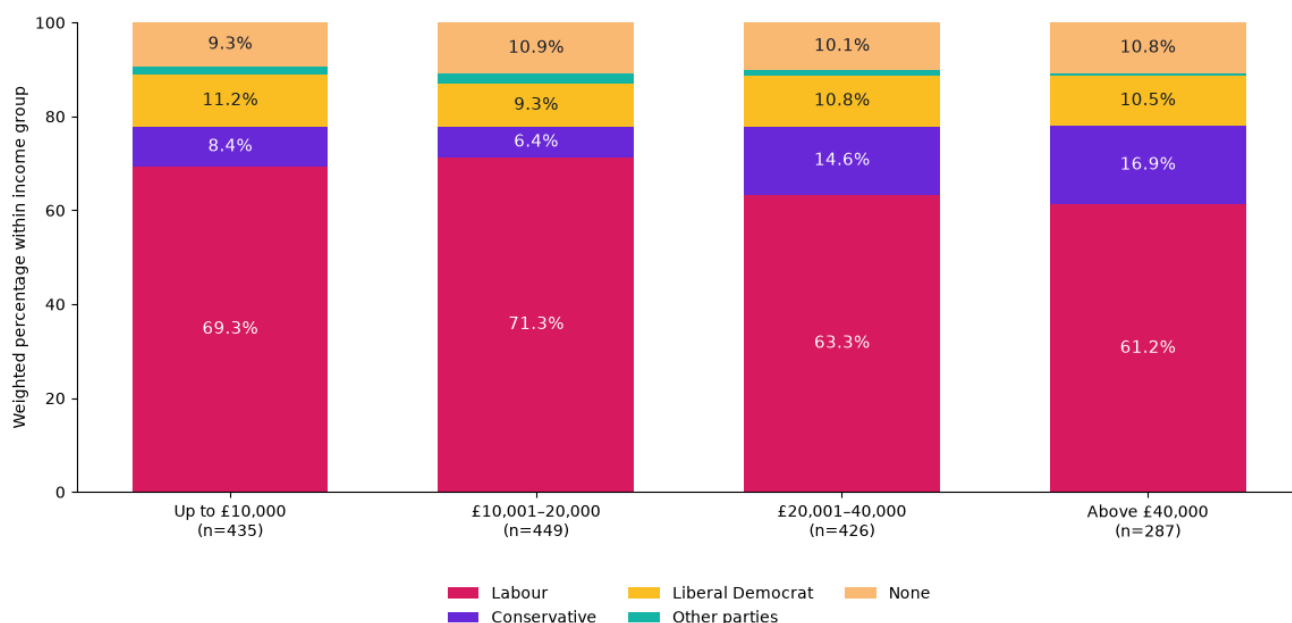


Both p-values are below 0.05; the smaller p-value does not identify a better grouping, df = degrees of freedom. Neither test adjusts for survey weights, sampling areas or strata. Final inference requires a survey-adjusted test. Effect sizes are unweighted. Changing income groups does not address bias from missing responses or establish causation.

We checked an alternative arrangement using three income groups. The p-value remained below 0.05 and the effect size remained small. This suggests our broad finding is not limited to the original four-group arrangement.

Exploring the distribution across four income groups however showed that there might be a difference in identification for the Labour and Conservative parties between low- and high-income groups.

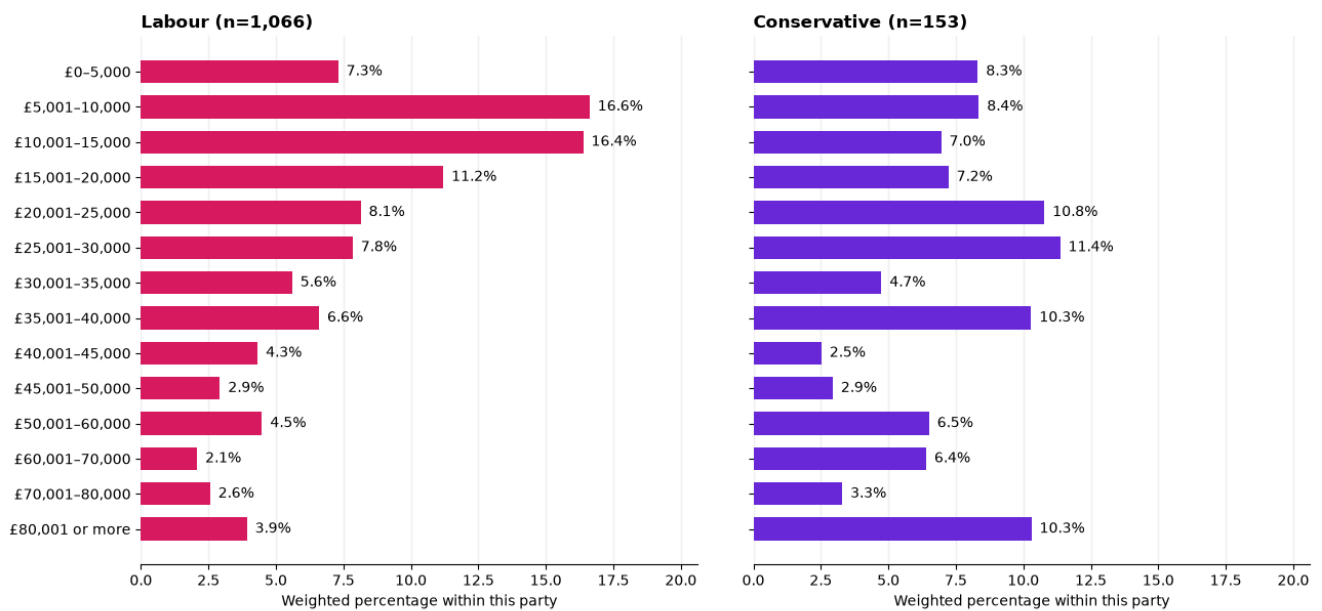
Party identification across household-income groups



EMBES 2010 | Complete cases: 1,597 | F2F ALL5 weights | Counts are unweighted. Income is annual household income before tax. Association does not establish causation.

2. Deeper analysis of the effect of income on Labour or Conservative identification

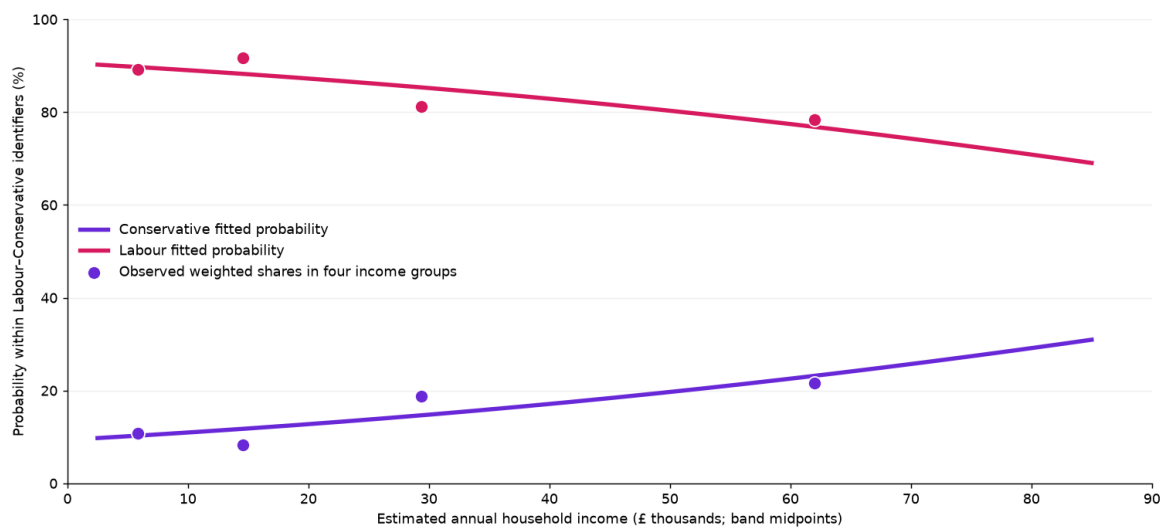
How household income is distributed within each party



EMBES 2010 | F2F ALL5 weights | n = unweighted respondents with valid income and party answers.
Each panel sums to 100%. Original income bands have unequal widths; bars show category shares, not income density.
This two-party comparison complements the all-party analysis; it does not establish causation.

Among those with valid income, **69.1% of Conservative identifiers reported household income above £20,000, compared with 48.5% of Labour identifiers**, using survey weights. This complements the main chart using the same data.

Income and identification within the two-party subset



EMBES 2010 | n=1,219 valid Labour/Conservative identifiers | F2F ALL5 weights.
Lines are complementary predictions from one logistic model; points are observed group shares, not confidence limits.
Highest income band assigned £85,000. Income-only fit; no demographic adjustment or survey-adjusted inference.

When transforming income from range baskets to median values to obtain a scale, we observed a **negative trend to identify with the Labour party as income increased**, and the **opposite effect for the identification to the Conservative party**, both quite significant ($p < 0.001$). This trend specifically showed that for every £10,000 increase in income, the probability to identify with the Labour party decreased by 2,5%, while the probability to identify with the Conservative party increased by 2,5%, which can be considered relatively small effects given the income scale from 0 to £95,000 here.

Our conclusion is therefore that **income is associated with party identification** in this analysis, but the **relationship is small. Labour remains dominant across income groups, however less in higher-income groups, where conservative identification is slightly increased.**

- Try to explain the impact of the result within the context of the data.

Overall our results in this context shows that household income is **only a weak predictor of party identification** for minorities.

We also have substantial missing income and have not adjusted for important characteristics such as ethnicity, age, education, or place of living, as it was not part of the task here. These findings should therefore be presented as exploratory, with **survey-adjusted testing as the next step.**

Within the context of the data, we suggest that party identification relies mostly on other predictors than household income. This result is most likely correlated with other major variables, rather than playing an actual role in party identification.

- Source code

The cleaned code is submitted in the form, but is also available on the following GitHub repository:

<https://github.com/ali112-akbar/CaseStudy>